

ECON 712: Applied Econometrics

Lecture 7,8: Linear Regression using Vector Spaces, part III

Linear Regression

- The concept of dropping a perpendicular is formalized by an orthogonal projection:

$$\mathbf{P}_X = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad , \quad \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{P}_X \mathbf{y}$$

- ▶ An orthogonal projection is a matrix that projects any vector to a point in the subspace $\mathcal{S}(\mathbf{X})$ that it is closest to it
 - ▶ $\mathcal{S}(\mathbf{X})$ is called the invariant subspace of $\mathbf{P}_X$ because the projection of any vector in $\mathcal{S}(\mathbf{X})$ using $\mathbf{P}_X$ leaves that vector unchanged
- We are also going to use a projection onto the orthogonal subspace of $\mathcal{S}(\mathbf{X})$, $\mathcal{S}^\perp$

$$\mathbf{M}_X = \mathbf{I} - \mathbf{P}_X \quad , \quad \hat{\mathbf{u}} = \mathbf{M}_X \mathbf{y}$$

Linear Regression

- Important properties of the projection matrices:

1. Idempotent:

$$\mathbf{P}_X \mathbf{P}_X = \mathbf{P}_X$$

$$\mathbf{M}_X \mathbf{M}_X = \mathbf{M}_X$$

2. Annihilation: $\mathbf{P}_X \mathbf{M}_X = \mathbf{M}_X \mathbf{P}_X = \mathbf{0}$

- Also, by Pythagorean Theorem (Sum Squares):

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_X \mathbf{y}\|^2 + \|\mathbf{M}_X \mathbf{y}\|^2$$

Linear Transformations

- Let $\mathbf{A}$ be a $k \times k$ matrix with full rank, and define $\mathbf{Z} = \mathbf{X} \mathbf{A}$ then :

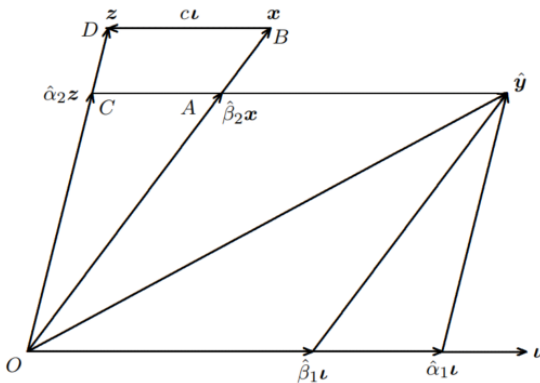
$$\mathbf{A} \hat{\alpha} = \hat{\beta}$$

where $\mathbf{X} \hat{\beta} = \mathbf{P}_X \mathbf{y}$, $\mathbf{Z} \hat{\alpha} = \mathbf{P}_Z \mathbf{y}$

- There are two cases (can be combined as in the Celsius/Fahrenheit example in ETM):
 - When the regressors are multiplied by a constant (e.g. change of units from cm to meters)
 - When a constant is added/subtracted from regressors

Linear Transformations

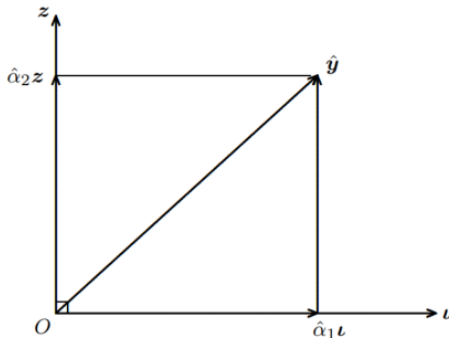
- Subtracting a constant ι_C from regressors when the constant term is included



Adding a constant does not affect the slope coefficient

Linear Transformations

- Subtracting a constant $\iota\bar{x}$ from regressors when the constant term is included



Orthogonal regressors may be omitted

Linear Transformations

- For a general regression where regressors are partitioned:
 $\mathbf{X} = [\mathbf{X}_1 \mathbf{X}_2]$, $\mathbf{X}_1$ is an $n \times k_1$, $\mathbf{X}_2$ is an $n \times k_2$; β_1 is $k_1 \times 1$, β_2 is $k_2 \times 1$,
and $k_1 + k_2 = k$

$$\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + u$$

the OLS/MM estimator has the following properties:

$$\hat{\mathbf{y}} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2$$

$$\hat{\mathbf{y}} = \mathbf{X}_1\hat{\alpha}_1 + \mathbf{Z}\hat{\beta}_2$$

where $\mathbf{Z} = \mathbf{X}_2 - \mathbf{X}_1 \mathbf{A}$ and $\mathbf{A}$ is a $k_1 \times k_1$ matrix

Linear Transformations

- In addition, if the transformation results in:

$$\langle \mathbf{X}_1, \mathbf{Z} \rangle = \mathbf{X}_1^T \mathbf{Z} = \mathbf{0}$$

then,

$$\hat{\mathbf{y}} = \mathbf{Z} \hat{\beta}_2$$

- This is the basis of the FWL theorem we studied in ECON 312:

► Let $\mathbf{Z} = \mathbf{X}_2 - \mathbf{P}_{\mathbf{X}_1} \mathbf{X}_2 = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$, then

$$\mathbf{P}_Z \mathbf{y} = \mathbf{Z} \hat{\beta}_2$$

- Note that residuals of this regression are not the same as the residuals of the original regression

FWL– with vector algebra

- Suppose that the OLS/MM estimator results in:

$$\mathbf{y} = \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}_2 \hat{\beta}_2 + \hat{\mathbf{u}}$$

and define $\mathbf{Z} = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$, $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1} \mathbf{y}$, then:

$$\mathbf{P}_Z \tilde{\mathbf{y}} = \mathbf{Z} \quad , \text{ and}$$

$$\mathbf{M}_Z \tilde{\mathbf{y}} = \hat{\mathbf{u}}$$

- This yields the following equation, where $\hat{\beta}_2$ denotes the OLS/MM estimator:

$$\tilde{\mathbf{y}} = \mathbf{Z} \hat{\beta}_2 + \hat{\mathbf{u}}$$

Goodness of Fit

- Recall that, by Pythagoras' Theorem:

$$TSS = \|\mathbf{y}\|^2 = \|\mathbf{P}_X \mathbf{y}\|^2 + \|\mathbf{M}_X \mathbf{y}\|^2 = ESS + SSR$$

- We use this result to obtain two goodness of fit measures:

$$R_u^2 = \frac{\|\mathbf{P}_X \mathbf{y}\|^2}{\|\mathbf{y}\|^2}$$

$$R_c^2 = \frac{\|\mathbf{P}_X \mathbf{M}_\iota \mathbf{y}\|^2}{\|\mathbf{M}_\iota \mathbf{y}\|^2}$$

- Note that the R^2 -centered is preferred, but it can only be interpreted when the constant term is included in the regression